EMBEDDED SYSTEMS LAB SETUP AND TEMPERATURE SENSOR INTERFACING

REPORT TASK-2

OBJECTIVE

The objective of this task is to set up an embedded development environment, interface a temperature sensor with an Arduino microcontroller, read sensor values, display them through the Serial Monitor, and perform an action when a predefined threshold is exceeded.

WORKING ENVIRONMENT

Environment Chosen: Wokwi Online Simulator

Microcontroller: Arduino UNO

Sensor: DHT11 Temperature & Humidity Sensor

Programming Language: Embedded C/C++ (Arduino)

COMPONENTS USED

- Arduino UNO
- DHT11 Temperature Sensor
- LED
- 220Ω Resistor
- Breadboard (Virtual)
- Jumper Wires

CIRCUIT CONNECTIONS

DHT11 Pin	Arduino Pin
VCC	5V
GND	GND
DATA	Digital Pin 2

LED Connections:

- **Positive (+)** → Digital Pin 13
- **Negative (-)** → GND through 220Ω resistor

WORKING PRINCIPLE

The DHT11 sensor measures the surrounding temperature and humidity.

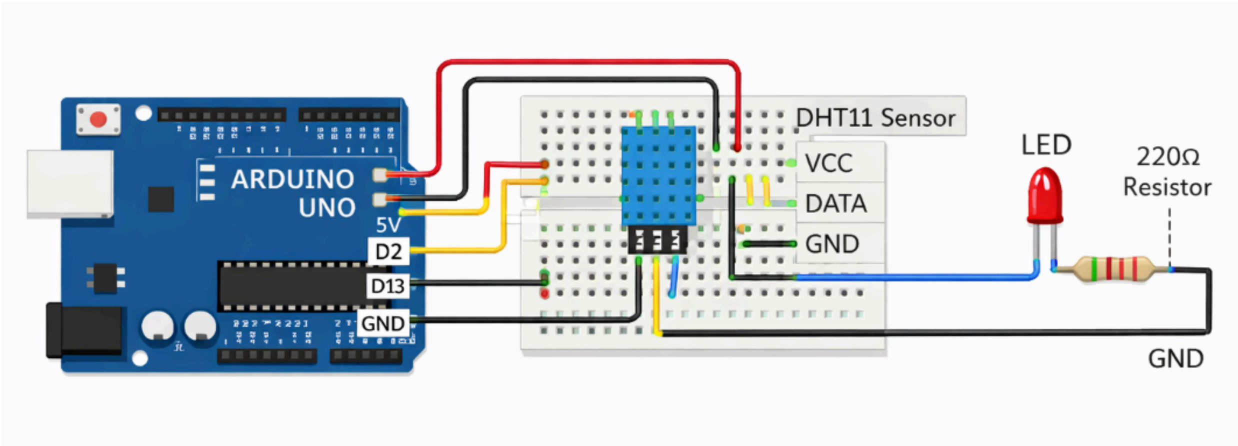
The Arduino continuously reads the temperature value from the sensor.

- If the temperature is less than or equal to **30°C**, the LED remains OFF.
- If the temperature becomes greater than **30°C**, the LED turns ON and an alert message is displayed in the Serial Monitor.

ARDIUNO PROGRAM

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

int led = 13;

void setup() {
  Serial.begin(9600);
  dht.begin();
  pinMode(led, OUTPUT);
}
void loop() {

  float temperature = dht.readTemperature();

  if (isnan(temperature)) {
    Serial.println("Sensor Error");
    return;
  }

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" °C");

  if (temperature > 30) {
    digitalWrite(led, HIGH);
    Serial.println("ALERT: High Temperature!");
  }
  else {
    digitalWrite(led, LOW);
  }

  delay(2000);
}
```

CIRCUIT DIAGRAM



OUTPUT

Example Serial Monitor Output:

```
Temperature: 27°C

Temperature: 29°C

Temperature: 31°C
ALERT: High Temperature!
```

LED Status:

- Temperature ≤ 30°C → LED OFF
- Temperature > 30°C → LED ON

OBSERVATIONS

- The DHT11 sensor successfully measured the temperature.
- Arduino displayed the readings in the Serial Monitor.
- The LED automatically turned ON when the temperature crossed the threshold value of **30°C**.
- The system responded correctly in real time.

APPLICATIONS

- Smart Home Automation
- Temperature Monitoring Systems
- Industrial Monitoring
- Weather Stations
- Greenhouse Monitoring
- IoT-based Environmental Monitoring

ADVANTAGES

- Simple and low-cost implementation.
- Easy sensor interfacing.
- Real-time monitoring.
- Expandable for IoT applications.
- Beginner-friendly project.

RESULT

The embedded development environment was successfully set up using the Wokwi simulator. The DHT11 temperature sensor was interfaced with the Arduino UNO, and the sensor readings were displayed on the Serial Monitor. The system successfully triggered an LED alert whenever the temperature exceeded **30°C**, achieving all the objectives of the experiment.